



Determining the Extent to Which a Deep Learning Algorithm Aligns with Las Vegas Predictions

A leakage-free Wide & Deep network predicts NBA games from pre-tip-off box-score form — and lands within a few points of the sharpest betting markets on earth.

Worked by : Eron Pano



Is 63% good? Depends what you stack it against.

- ▶ This Model:
 - ▶ Pre-Game form only
 - ▶ No Injuries, Lineups, or betting signal
 - ▶ Around 63% Accuracy
- ▶ Las Vegas Model:
 - ▶ Rooms full of quants
 - ▶ Insider Injury information
 - ▶ Betting information
 - ▶ Most accurate NBA forecast at 68-70% accuracy

Single-game prediction is deceptively hard

- ▶ Home win – base rate:
 - ▶ 54.4%
- ▶ Las Vegas Ceiling:
 - ▶ 67%
- ▶ Useful models:
 - ▶ Climb from 54.4% to 67% accuracy



Ten seasons, rebuilt to respect time

- 11876 games
- 64 leakage-free features
- 10 seasons
- 13.5k model parameters
- Workflow:
 - Step 1: Fetch stats from NBA API
 - Step 2: Gather one game per row
 - Step 3: Engineer rolling L5 and L10 form
 - Step 4: Time-based split

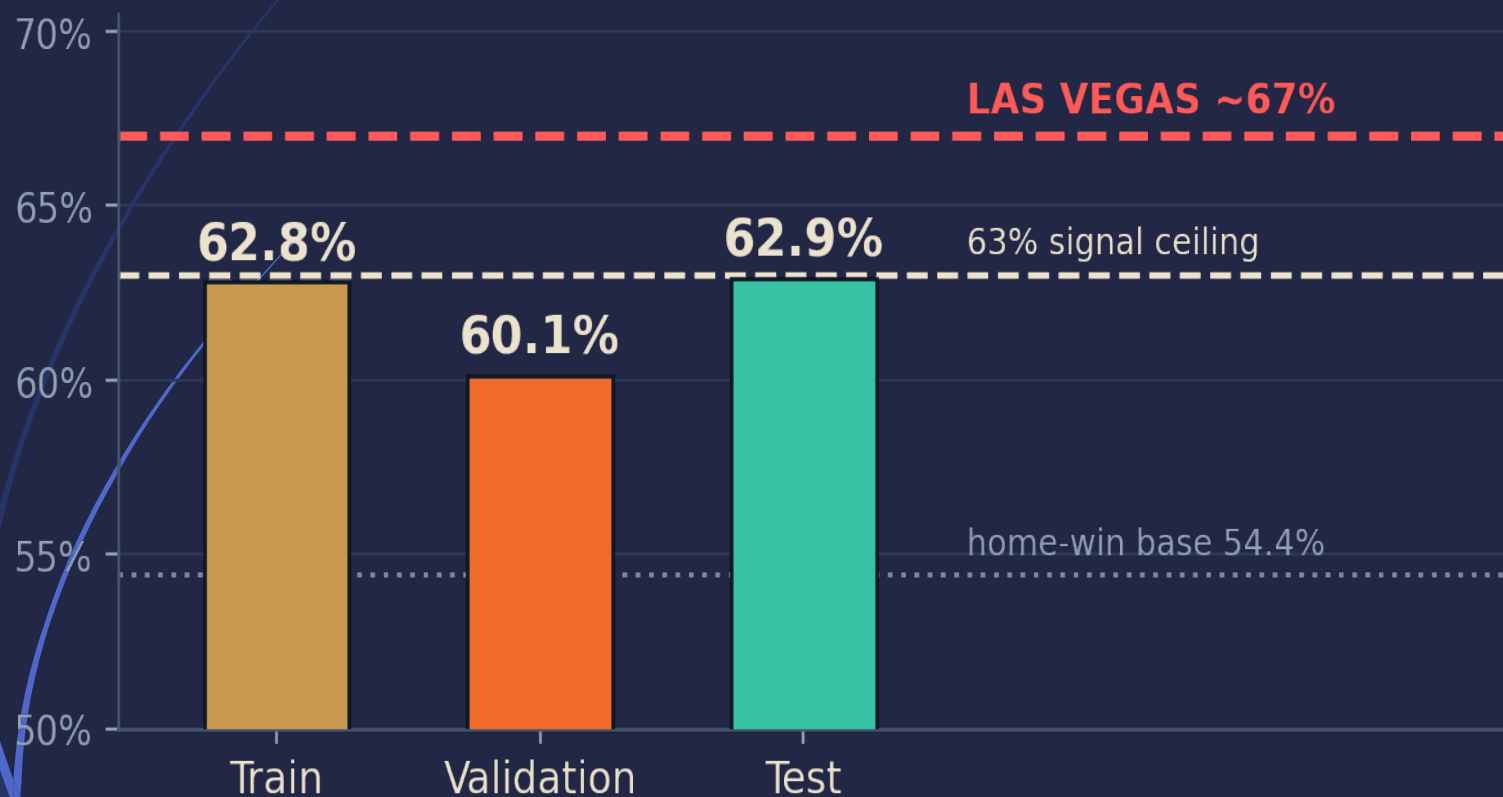
An 87% model was lying. Fixing it gave 63%.

- Received an accuracy of 87-88% because of data leakage
- Tested multiple models on new data
- Decided on the Wide and Deep model
- 62.9% is the accuracy received
- Best possible accuracy on leakage-free data

Wide & Deep: memorize and generalize

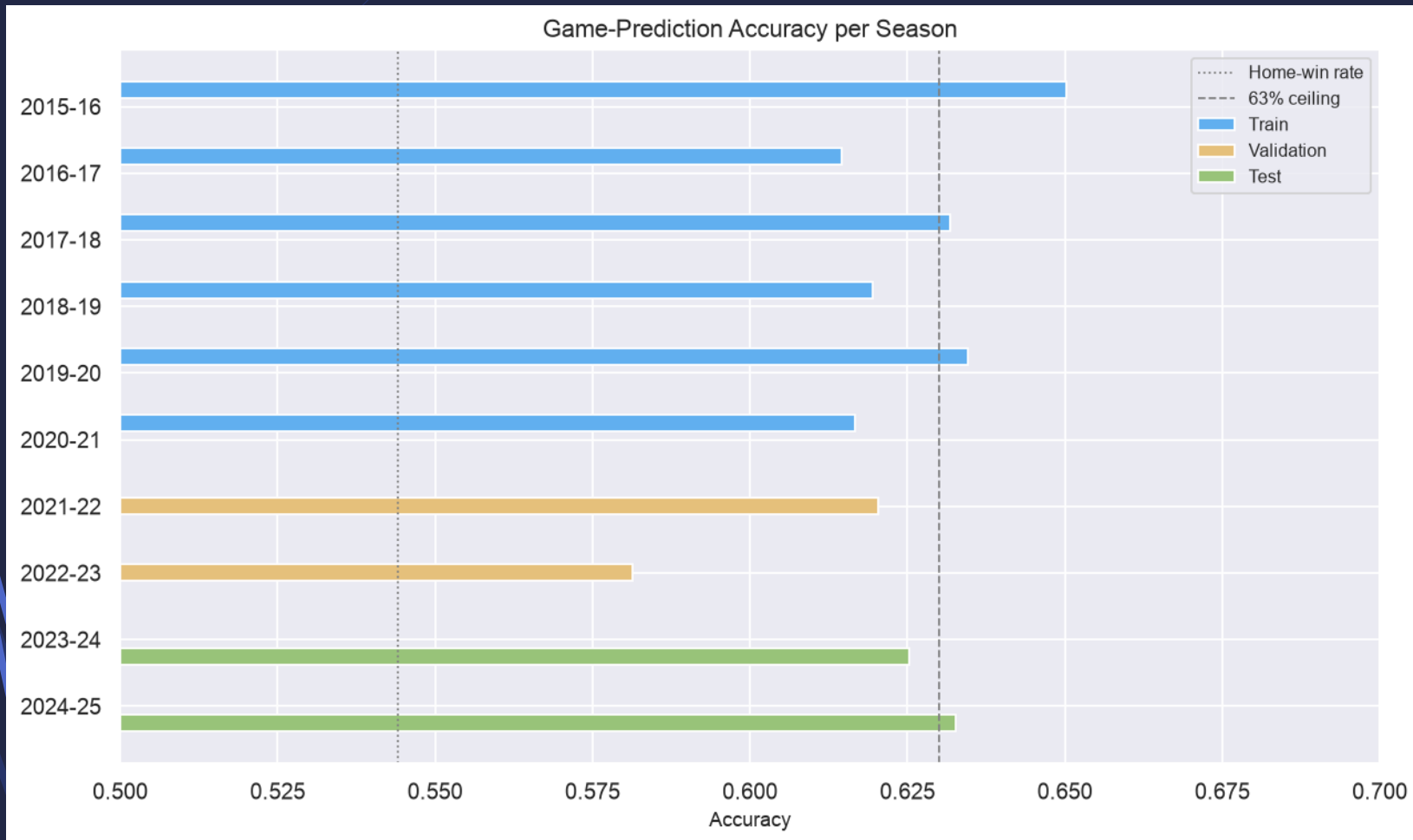
- ▶ 64 Features
- ▶ Wide – Linear Branch
 - ▶ memorizes direct signals (16 units)
- ▶ DEEP – Learns Interactions
 - ▶ $96 \rightarrow 48 \rightarrow 24$, ReLU + BatchNorm
 - ▶ dropout 0.5 / 0.4 / 0.3

Every split lands at the ceiling — just shy of Vegas



- ▶ Test Accuracy:
 - ▶ 62.9%
- ▶ Test ROC-AUC:
 - ▶ 0.676
- ▶ Train Accuracy:
 - ▶ 62.8%
- ▶ Gap to Vegas:
 - ▶ 4 pts

Accuracy holds season to season

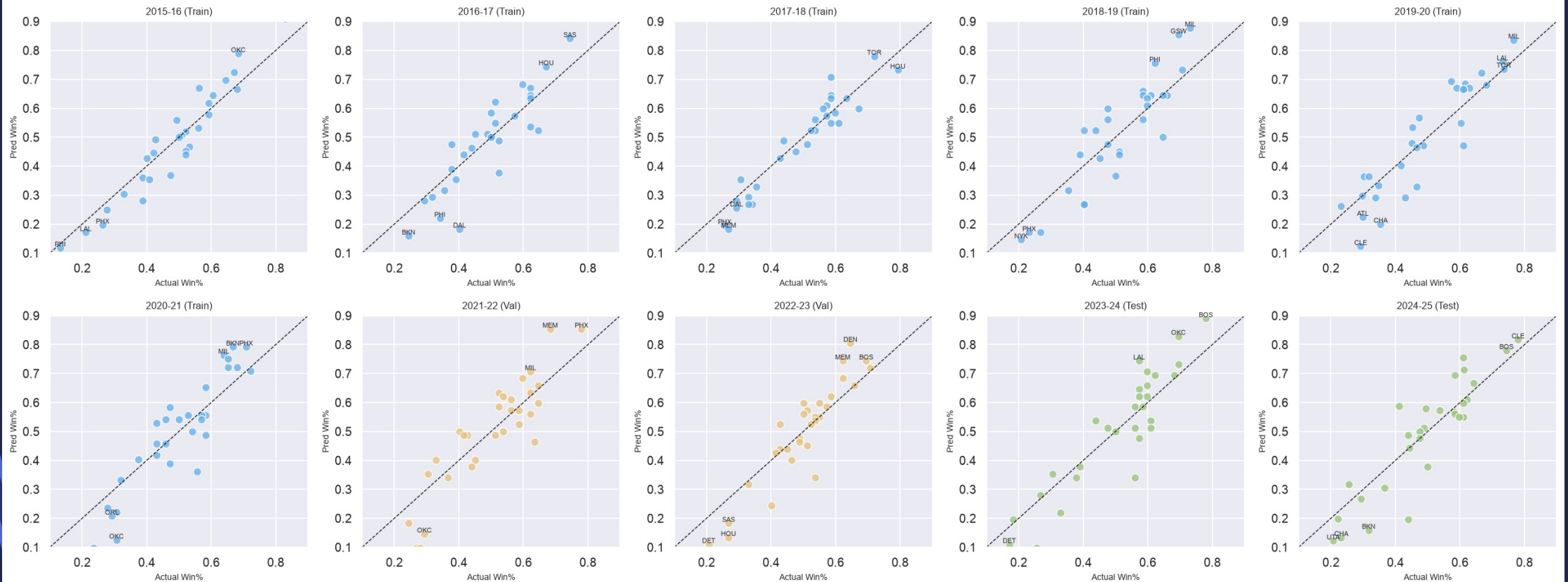


Train $\approx$ Test

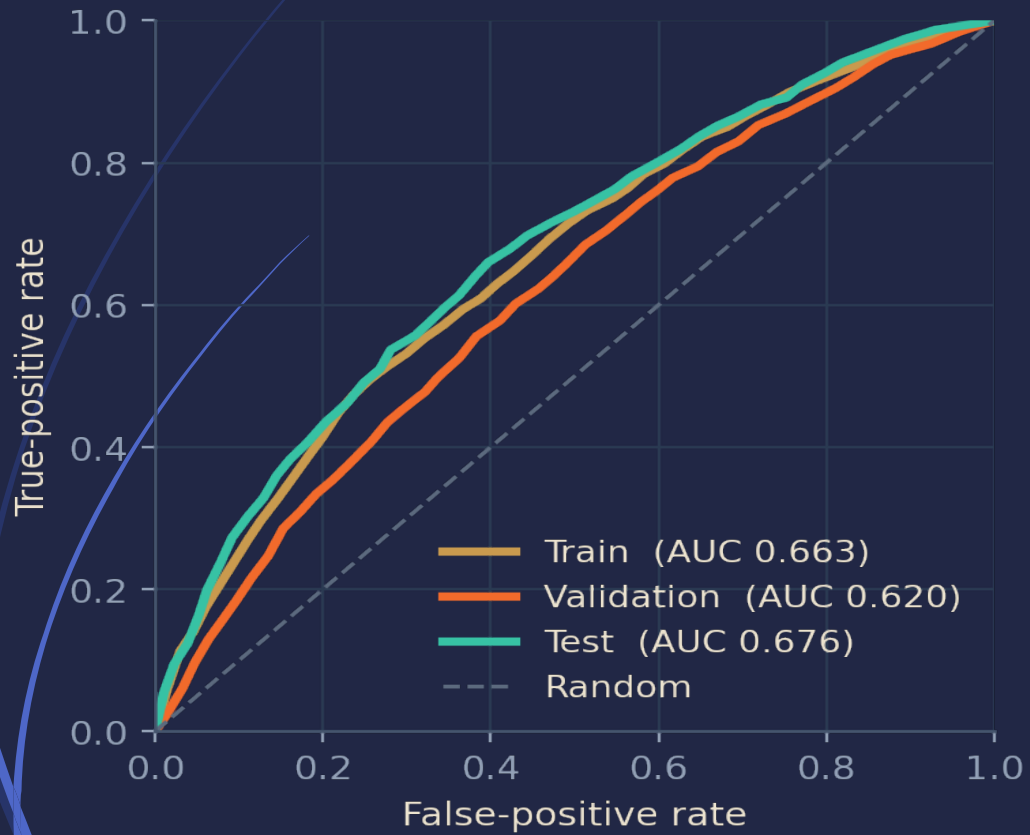
The held-out 2023–24 and 2024–25 seasons score right alongside the training seasons. The model generalizes to years it never saw rather than memorizing the past.

Predicted win% tracks reality

Predicted vs Actual Win % by Season



Well above chance in every split



- ▶ Test AUC: 0.676

- ▶ Best of the three splits — performance does not decay on unseen data.

- ▶ Train AUC: 0.663

- ▶ Close to test: the model is not memorizing.

- ▶ Validation AUC 0.620

- ▶ The noisy 2022–23 season pulls this down slightly.

Three reasons 63% is exactly right

- **63% is the real limit**

- Bigger, deeper, less-regularized models did not move the number. That's the signal ceiling of the data, not a tuning failure.

- **A high score is a red flag**

- If this returned 90%, the right response is suspicion. Honest features cap out near the market — anything above means leakage.

- **Train $\approx$ Test**

- 62.8% train vs 62.9% test, within a tenth of a point. The model generalizes; dropout and weight decay keep it honest.



Conclusion:

a model can hang with Vegas

- ▶ A leakage-free Wide & Deep network reaches 62.9% on unseen seasons — a few points under the ~67% market line, with the gap fully explained by the injury and lineup information Vegas prices in and the model deliberately omits.
- ▶ **Real contribution** Diagnosing and removing temporal leakage — not chasing a vanity number.
- ▶ **Honest by design** 62.9% that survives deployment beats a brittle 90% that collapses.
- ▶ **Next** Sequence models, injury & lineup data, and calibration against closing lines.